

SAM A. BNMORAAD

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New York City, NY

COMPUTER SCIENTIST

I am a results-driven software engineer with a passion for backend systems, data optimization, and scalable solutions. My background spans financial institutions, postal logistics, and academic research, with a consistent record of improving system efficiency, ensuring compliance, and enabling better decision-making through data.

Areas of Expertise

Backend Development • API Optimization • Microservices Architecture • Data Analysis • PostgreSQL Optimization • OAuth2/JWT • Docker & Kubernetes • Scientific Computing • Medical Software Tools • Research & Technical Writing • Multilingual Communication (English, French, Arabic)

EDUCATION

Bachelor of Science in Computer Science | Columbia University – New York City, NY

May 2027

PROFESSIONAL WORK HISTORY

Backend Software Developer | Zitouna Bank – Remote

January 2024 – May 2024

- Engineered and optimized RESTful APIs using Python and Django, cutting response times by 30% and boosting transaction throughput by 25%
- Partnered with database teams to manage and tune PostgreSQL performance for real-time, high-volume financial data
- Ensured backend infrastructure aligned with security standards and financial regulations, resulting in a 100% compliance rate during audits

Software Engineer | Tunisian Postal Office – Remote

April 2023 – September 2024

- Led optimization of parcel tracking APIs and contributed to a microservices-based system redesign for higher scalability and modularity
- Integrated OAuth2 and JWT for secure access management, reducing unauthorized access incidents
- Reduced API latency by 40% and improved tracking accuracy by 25%

Freelance Software Developer | Fiverr – Remote

November 2023 – May 2024

- Delivered customized software and cybersecurity solutions for 50+ clients, including API integration, mobile/web app development, and performance tuning

Finite Element Analysis Research | Lehigh Collaboration

November 2021 – December 2021

- Designed C++ plugins for ABAQUS to simulate cranial bone remodeling post-trauma.
- [Recognized as a Peer Reviewer for Annals of Internal Medicine](#) for outstanding contribution.

Freelance Software Developer | Fiverr – Remote

November 2023 – May 2024

- Developed an automated Python benchmarking toolkit for evaluating large language model (LLM) responses in LGBTQ+ suicide prevention, contributing to a peer-reviewed publication on ethical alignment and emotional intelligence in AI-driven mental health chatbots.

TECHNICAL SKILLS

Data Programming Languages: Python, C++, Java, JavaScript, SQL, HTML/CSS, Ruby

Backend & DevOps: Django, RESTful APIs, OAuth2, JWT, Docker, Kubernetes

Data Analysis & Databases: PostgreSQL, ABAQUS, ANSYS, MATLAB, R

Scientific & Analytical Tools: Finite Element Modeling, Simulation Software

Web & App Development: API Integration, Web Optimization, Microservices Architecture

Languages: English (Native), French (Fluent), Arabic (Fluent)